

WANDO, Korea, Republic of

WMO: 471700

Lat: 34.3958N

Lon: 126.7019E

Elev: 120

StdP: 14.63

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	24.7	27.2	8.3	8.4	33.0	11.9	10.1	33.0	29.9	34.2	27.5	33.9	10.6	340	0.432

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.2	89.0	79.5	87.0	78.6	84.9	77.5	81.1	86.2	79.8	85.2	78.6	83.8	8.4	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.8	155.5	84.4	78.2	147.1	83.3	77.1	141.5	82.2	45.0	86.6	43.5	85.5	42.3	84.1	88.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.0	22.4	19.4	21.8	92.5	3.2	3.2	19.5	94.8	17.6	96.7	15.8	98.5	13.4	100.8
			19.2	82.2	2.7	2.3	17.2	83.9	15.6	85.2	14.1	86.5	12.1	88.2
			DB											
			WB											

Monthly Climatic Design Conditions

			Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
			(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	58.3	36.8	39.4	46.3	55.7	64.3	70.8	77.4	79.5	72.7	63.2	51.8	41.0	(6)
(7)		DBStd	15.41	5.43	5.91	5.72	5.04	4.17	3.21	4.28	3.74	4.11	4.96	6.27	5.81	(7)
(8)		HDD50	1201	408	301	143	11	0	0	0	0	0	1	53	284	(8)
(9)		HDD65	3753	873	718	580	281	65	2	0	0	1	91	397	745	(9)
(10)		CDD50	4248	1	3	28	181	442	625	849	915	682	411	107	4	(10)
(11)		CDD65	1324	0	0	0	1	42	177	384	450	233	36	1	0	(11)
(12)		CDH74	8832	0	0	0	3	169	600	2917	4007	1070	66	0	0	(12)
(13)		CDH80	2201	0	0	0	0	14	60	766	1194	165	2	0	0	(13)
(14)	Wind	WSAvg	7.8	9.5	9.2	9.0	8.5	7.6	6.8	7.2	7.0	6.4	6.7	7.6	8.5	(14)
(15)	Precipitation	PrecAvg	58.3	1.3	2.1	3.8	5.2	5.3	7.4	10.2	10.2	6.4	2.4	2.6	1.5	(15)
(16)		PrecMax	88.9	2.2	5.8	7.1	11.2	10.6	17.4	22.2	24.4	21.5	7.3	9.4	3.4	(16)
(17)		PrecMin	35.2	0.2	0.2	0.8	0.9	1.9	1.0	1.6	1.2	0.9	0.2	0.1	0.3	(17)
(18)		PrecStd	12.8	0.6	1.3	1.7	2.6	2.1	4.1	5.0	5.8	5.5	1.7	2.2	0.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.7	60.2	65.9	73.7	81.9	84.4	91.4	92.8	86.9	78.7	70.0	58.8	(19)
MCWB			49.1	55.1	54.4	58.7	64.3	71.6	80.4	80.4	76.6	68.2	62.7	52.6	(20)	
2%		DB	50.2	55.3	61.8	70.4	78.3	81.7	88.8	89.8	83.9	75.8	66.3	54.8	(21)	
		MCWB	44.6	49.5	52.7	58.4	63.1	69.8	79.3	79.9	74.3	65.4	59.5	49.0	(22)	
5%		DB	47.4	51.8	58.6	67.5	75.4	79.7	86.6	87.9	81.7	73.6	63.6	52.2	(23)	
		MCWB	42.4	45.1	50.8	57.0	62.1	69.7	78.8	79.3	72.8	64.8	57.5	47.0	(24)	
10%		DB	44.8	48.5	55.9	64.7	72.8	77.6	84.3	85.9	79.7	71.1	61.0	49.7	(25)	
		MCWB	39.8	42.6	49.2	55.8	61.6	69.2	77.8	78.3	72.1	62.8	55.5	44.6	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.6	57.4	59.0	65.9	70.8	76.5	83.9	83.5	79.0	73.6	64.8	54.5	(27)
MCD B			53.6	59.4	62.2	69.4	73.8	79.8	87.1	88.5	84.4	75.7	67.6	57.1	(28)	
2%		WB	46.0	51.2	55.6	62.7	67.4	74.3	81.2	81.5	77.1	69.3	61.7	50.7	(29)	
		MCD B	49.1	54.2	58.9	66.4	72.7	77.8	85.5	87.1	81.5	72.7	64.8	53.9	(30)	
5%		WB	43.1	46.3	52.8	60.1	65.7	72.7	79.5	80.3	75.6	66.5	59.3	47.8	(31)	
		MCD B	46.5	50.6	56.9	64.3	71.5	76.4	84.9	85.9	79.5	70.7	62.4	51.2	(32)	
10%		WB	40.4	43.5	50.0	57.7	64.0	71.2	78.3	79.2	73.9	64.4	56.8	45.2	(33)	
		MCD B	44.1	47.4	55.0	62.1	69.7	75.2	83.5	84.5	77.6	69.7	60.2	49.2	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.2	12.4	14.2	14.8	14.3	11.5	9.6	10.2	11.7	13.4	12.5	11.2	(35)
MCD BR			14.2	17.3	18.8	19.2	19.3	16.1	12.9	13.2	14.1	15.7	15.0	14.9	(36)	
MCWBR			10.3	12.1	12.3	11.0	9.8	7.9	6.2	5.9	6.7	8.1	9.6	10.8	(37)	
5% WB		MCD BR	13.1	15.4	15.8	14.6	14.8	10.5	11.7	11.4	11.0	12.5	12.5	13.4	(38)	
		MCWBR	10.9	12.6	12.6	11.3	9.2	6.7	6.8	6.0	6.4	8.5	9.8	11.2	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.448	0.491	0.554	0.568	0.583	0.631	0.599	0.552	0.501	0.469	0.456	0.444	(40)	
taud		1.968	1.896	1.768	1.789	1.806	1.782	1.912	2.040	2.115	2.067	2.024	1.987	(41)		
Ebn at Noon		227	233	229	235	233	221	228	236	242	237	223	219	(42)		
Edh at Noon		46	55	67	69	69	70	61	53	47	45	43	43	(43)		
(44)	All-Sky Solar Radiation	RadAvg	775	1017	1377	1603	1728	1509	1480	1538	1334	1189	852	698	(44)	
(45)		RadStd	63	93	117	130	196	152	207	194	133	111	86	42	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	-2.66	N/A	N/A	N/A	N/A	N/A	+182	+101
(47)	Regional (25 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page